Advances in SAM-Based Medical Image Segmentation

A Survey of Agentic, Weak-Supervision & Preference Optimization Approaches

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Outline

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Need for generalizable medical segmentation

02 **MedSAM-Agent**

Agentic multi-turn reinforcement learning

03 **All-in-SAM**

Weak annotation → pixel-wise segmentation

04 **MAPO**

Model-agnostic preference optimization

05 **Comparative Analysis**

Side-by-side: all three methods

06 **Discussion & Conclusions**

Key findings, limitations, future work

Background & Motivation

THE CHALLENGE

- U-Net / nnU-Net → task-specific retraining
- SAM: strong zero-shot, weak on medical
- Domain gap: modality noise, low tissue contrast
- Manual prompts → not scalable clinically
- Annotation cost = critical AI bottleneck

THREE RESEARCH DIRECTIONS

MedSAM-Agent

Agentic RL · autonomous multi-turn

All-in-SAM

Label-efficient · weak annotation bootstrap

MAPO

Architecture-agnostic · preference optimization

Segmentation Paradigms — The Problem Space

(a) SAM-based Models

- Expert prompts per image
- Manual point/box required
- ✗ Scalable ✓ Precise

(b) MLLM-driven Models

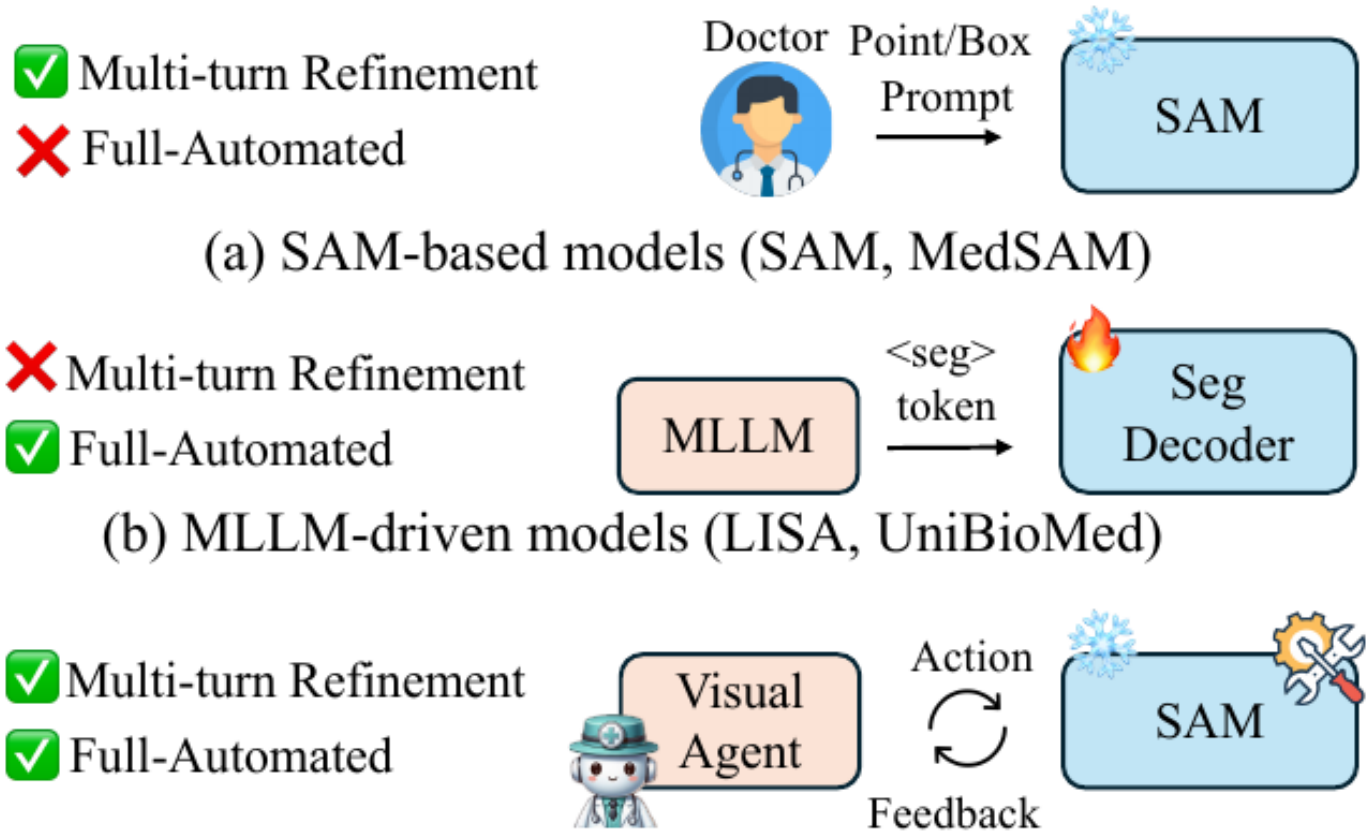
- <seg> tokens + pixel decoders
- Automated generation
- ✓ Scalable ✗ Spatially precise

(c) MedSAM-Agent

- Visual agent + iterative SAM tools
- Multi-turn: bbox → point → stop
- ✓ Scalable ✓ Precise

Source: Liu et al., MedSAM-Agent — arXiv:2602.03320, 2026

Figure 1: Comparison of Segmentation Paradigms



MedSAM-Agent: Framework Overview

MDP Formulation

- State s_t · Actions: bbox / point / stop
- SAM as tool, MLLM as agent
- Observation = updated mask

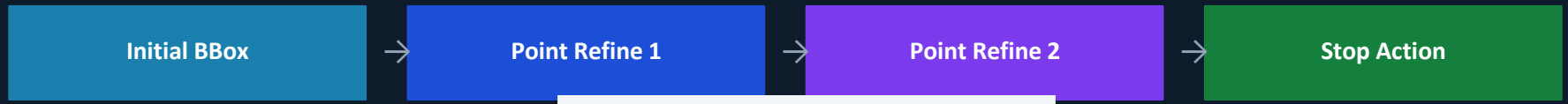
Hybrid Trajectory Gen.

- Box-to-Point + Sequential-Click
- IoU progress gate $\tau = 0.04$
- 449K training samples

2-Stage Training

- Stage 1: SFT cold-start
- Stage 2: GRPO-RL
- Composite process reward

MULTI-TURN INTERACTION FLOW



Liu et al., arXiv:2602.03320, 2026

MedSAM-Agent: Process Reward Design

$$R_{total} = w_1 \cdot R_{fmt} + w_2 \cdot \text{clip}\left(R_{qual} + \lambda_1 \cdot R_{imp} - \lambda_2 \cdot R_{over} - \lambda_3 \cdot R_{cost},\ 0,\ 1 \right)$$

R_fmt

Format adherence · Tool call + stop action · each = 0.5

R_qual

Segmentation quality · w_iou · IoU_final + w_dice · Dice_final

R_imp ★

Progressive improvement · $\Sigma \max(0, IoU_t - IoU_{t-1})$ · MOST CRITICAL

R_over

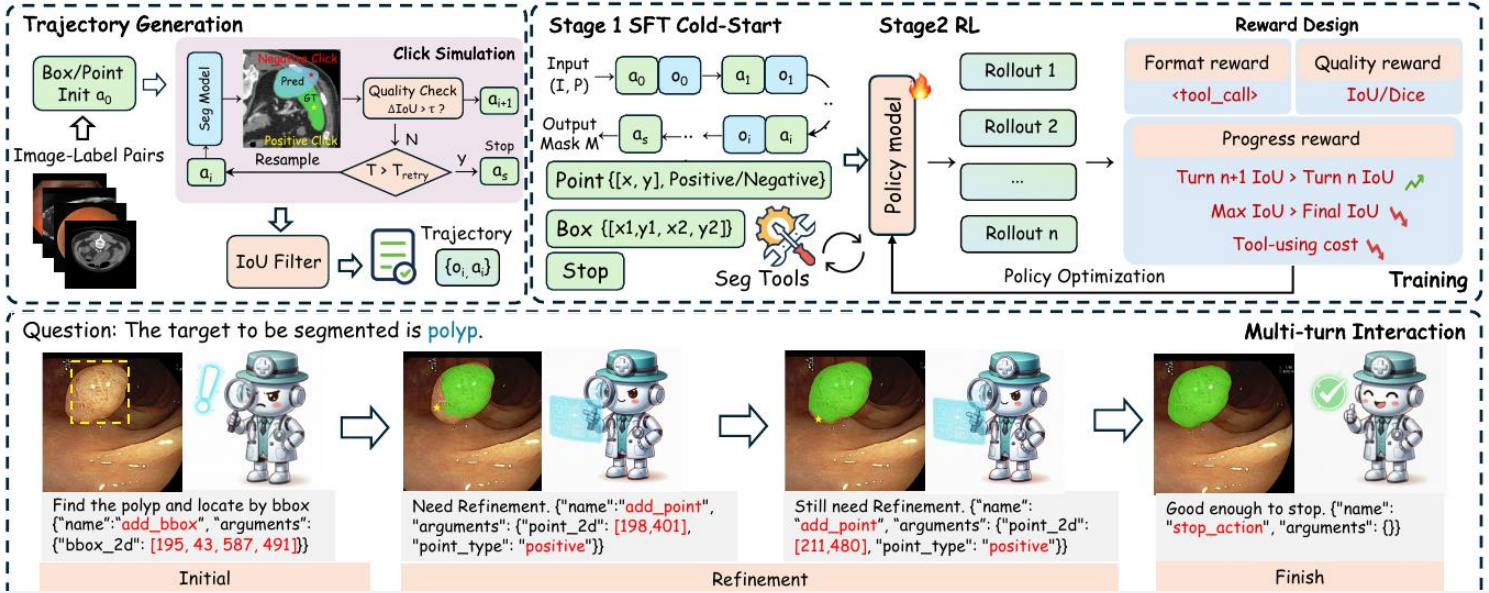
Overshoot penalty · IoU_max − IoU_final · prevents quality drop

R_cost

Tool-use efficiency · Linear penalty on trajectory length

★ Removing R_imp: Dice 0.794 → 0.772 (−0.022) — largest single drop

Figure 2: MedSAM-Agent Overview



MedSAM-Agent: Results — 6 Modalities, 21 Datasets

0.794	0.705	2.11	21
Avg Dice	Avg IoU	Avg Turns	Datasets

Method	CT	MRI	X-Ray	US	Avg Dice
SAM2-Point*	0.610	0.488	0.546	0.528	0.532
SAM2-Box*	0.837	0.838	0.827	0.871	0.835
MedSAM-Box*	0.847	0.854	0.897	0.890	0.876
UniBiomed	0.724	0.807	0.817	0.736	0.776
Seg-R1	0.099	0.126	0.417	0.490	0.353
Ours (MedSAM2) ✓	**0.720**	**0.793**	**0.833**	**0.793**	**0.794**

** Interactive methods using ground-truth prompts (upper bound)*

MedSAM-Agent: Ablation Study

R_imp most critical

Removing R_imp: 0.794 → 0.772 (−0.022), largest single drop

Hybrid prompting wins

Sequential: 0.623 · Box-to-Point: 0.649 · Hybrid: 0.705 IoU

SFT essential

RL alone fails CT/MRI — initial grounding problem

51% fewer turns

Full model: 2.11 avg turns vs 4.34 base — major efficiency gain

Liu et al., MedSAM-Agent — arXiv:2602.03320, 2026

All-in-SAM: Weak Annotation → Pixel-wise Segmentation

Stage 1
SAM-Empowered Annotation

- Input: bounding boxes only
- Frozen SAM → pseudo pixel masks
- SAM Dice on MoNuSeg (box): 0.883

Stage 2
Prompt-Based Fine-Tuning

- SAM-Adapter: high-freq features → ViT
- Feature fusion: $\tilde{z}^n = z^n + f\phi(h_HF(I))$
- Inference: zero manual prompts

Cui et al., All-in-SAM — arXiv:2307.00290, 2023

All-in-SAM: Results on MoNuSeg

4% data + bbox labels → 80.99% Dice ≈ nnU-Net on 100% complete annotations (82.44%)

Label	Method	Data	Dice (%)	IoU (%)
Complete	nnUNet	All	82.44	69.76
Complete	nnUNet	4%	79.20	65.40
Complete	Proposed ★	All	82.54	69.74
Complete	Proposed ★	4%	81.34	68.10
Weak	nnUNet	All	82.12	69.35
Weak	nnUNet	4%	79.13	65.27
Weak	Proposed ★	All	82.46	69.73
Weak	Proposed ★	4%	80.99	67.60

★ Best result in category | Cui et al., arXiv:2307.00290, 2023

MAPO: Model-Agnostic Preference Optimization

Stage 1

Warm-up SFT

- Dice + CE supervised training
- Establishes baseline capability

Stage 2

Dropout Preference Gen.

- Rates $d_1 \ldots d_m$ applied
- Best $\hat{y}^+$ and worst $\hat{y}^-$ selected

Stage 3

DPO Optimization

- $L = \lambda(L_{Dice}+L_{CE}) + L_{DPO}$
- $\beta=0.1$, $\lambda=0.5$

Stage 4

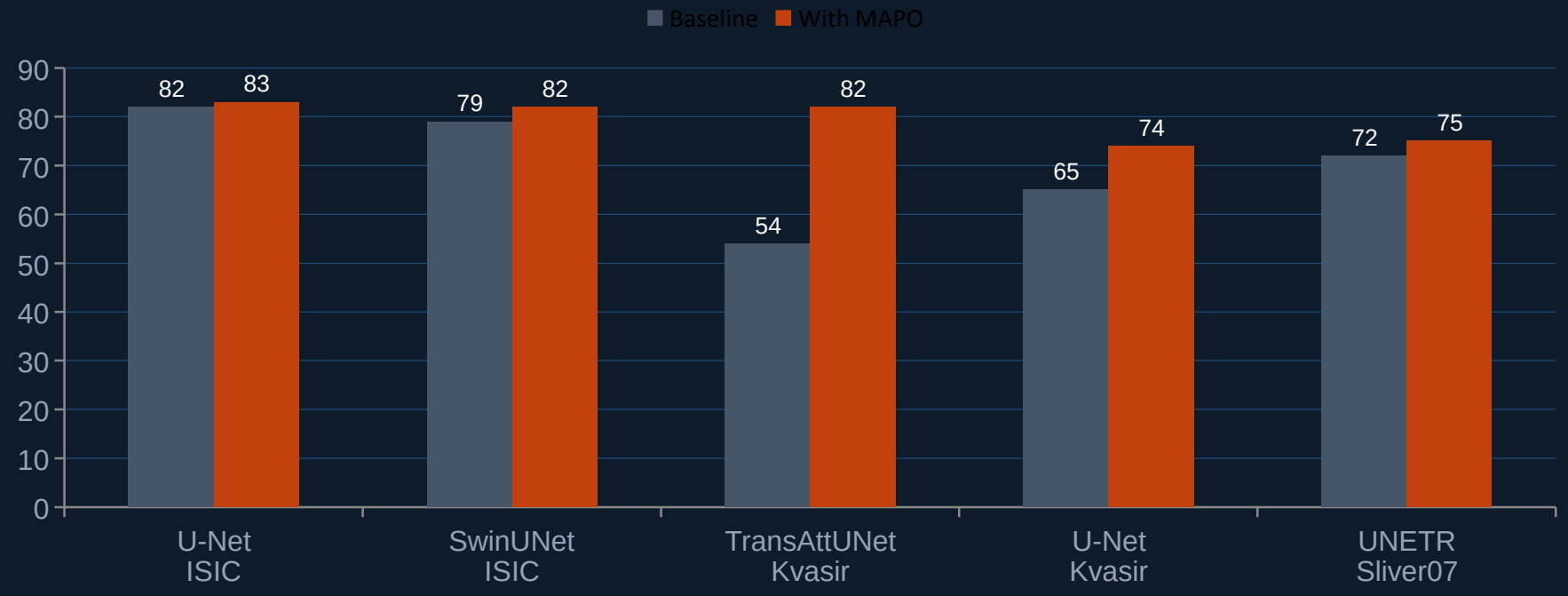
Online Update

- Regenerate pairs every 50 epochs (2D)
- Every 200 epochs (3D)

Nam et al., MAPO — arXiv:2512.15009, 2024

MAPO: Results — 2D Segmentation

+28.93% Dice improvement on Kvasir (TransAttUNet) — MAPO improves all tested architectures



MAPO: Results — 3D Segmentation

Consistent Dice & ASD improvement across CNN, ViT, and hybrid architectures on 3 datasets

Method	Parse Dice	Parse ASD	Sliver Dice	Sliver ASD	Luna Dice	Luna ASD
VNet (CNN)	66.25	13.20	71.23	30.79	96.98	2.18
VNet+MAPO	**68.79**	12.90↓	**73.40**	27.94↓	**96.99**	1.56↓
DynUNet (CNN)	72.95	17.78	62.90	27.69	95.96	5.10
DynUNet+MAPO	**74.20**	15.20↓	65.25	21.22↓	94.65	4.90↓
DFormer (ViT)	65.57	15.71	67.60	26.87	96.98	3.93
DFormer+MAPO	**67.60**	13.11↓	69.50	21.11↓	97.02	3.90↓
UNETR (Hybrid)	60.01	18.15	72.31	26.27	96.47	1.32
UNETR+MAPO ★	**62.76**	15.47↓	**74.93**	21.35↓	97.01	4.26

★ Best on Sliver07 | Nam et al., arXiv:2512.15009, 2024

Comparative Analysis: Three Complementary Approaches

MedSAM-Agent

Liu et al., 2026

Paradigm	MLLM agent → SAM tools
Training	SFT cold-start + GRPO-RL
Input	Text prompt only
Process	Multi-turn: bbox → point → stop
Result	Avg Dice 0.794 / IoU 0.705
Limits	2D only · inference latency

All-in-SAM

Cui et al., 2023

Paradigm	SAM as annotator + backbone
Training	SAM-Adapter on pseudo labels
Input	Bounding-box weak labels
Process	Single-pass inference, no prompts
Result	Dice 82.54% (100%) · 80.99% (4%)
Limits	Only MoNuSeg validated

MAPO

Nam et al., 2024

Paradigm	Dropout-driven DPO any arch
Training	Warm-up SFT + online DPO
Input	Full pixel labels for warm-up
Process	Single-pass · preference-based
Result	+0.1% to +28.9% Dice (8 datasets)
Limits	Warm-up needs pixel labels

Key Findings

01

Multi-turn > Single-turn

- MedSAM-Agent surpasses oracle single-turn box prompts (all 6 modalities)
- R_imp per-step supervision > terminal-only rewards

02

Weak Labels ≈ Strong Labels

- 4% data + bbox → 80.99% Dice vs nnU-Net 82.44% (100% complete)
- SAM edge detection is the critical enabler

03

DPO as Universal Regularizer

- MAPO improves all 7 tested architectures
- Largest gains (+28.9%) on unstable baselines

04

Three Orthogonal Contributions

- Autonomy · Annotation efficiency · Training alignment → independent
- Unified pipeline = most promising future direction

Limitations & Future Directions

LIMITATIONS

MedSAM-Agent

2D only · multi-turn latency · SAM backend required

All-in-SAM

MoNuSeg only validated · Inherits SAM low-contrast weakness

MAPO

Pixel labels needed for warm-up · Online regen adds compute overhead

FUTURE DIRECTIONS

- 3D / volumetric agentic segmentation + cross-slice memory
- Unified pipeline: MedSAM-Agent + All-in-SAM + MAPO
- MAPO DPO applied to multi-turn agent trajectories
- Early-exit & KV-cache → real-time clinical deployment
- Large-scale weak-label pretraining → universal medical SAM

Key References

[1] S. Liu et al. MedSAM-Agent: Empowering Interactive Medical Image Segmentation with Multiturn Agentic Reinforcement Learning. arXiv:2602.03320, 2026

[2] C. Cui et al. All-in-SAM: From Weak Annotation to Pixel-wise Nuclei Segmentation with Prompt-based Finetuning. arXiv:2307.00290, 2023

[3] Y. Nam et al. Model Agnostic Preference Optimization for Medical Image Segmentation. arXiv:2512.15009, 2024

[4] A. Kirillov et al. Segment Anything. CVPR, 2023

[5] F. Isensee et al. nnU-Net: A Self-configuring Method for Deep Learning-based Biomedical Image Segmentation. Nature Methods, 18(2): 203–211, 2021

Thank You

MedSAM-Agent —	Agentic RL · Avg Dice 0.794 · 21 datasets · 6 modalities
All-in-SAM —	Weak annotation · 80.99% Dice at 4% data · No inference prompt
MAPO —	Architecture-agnostic DPO · Up to +28.9% Dice · 2D & 3D

Three orthogonal contributions → annotation-efficient, autonomously-refined, preference-aligned clinical medical AI